

4 Chapter Four: Prototype Experimentation

4.1 Introduction

This chapter discusses the way that the prototype has been evaluated. These evaluations are important in assessing the viability and validity of the prototype's usage. The remainder of the chapter is structured as follows: section 4.2 will discuss the different changes that will be made to the PDFs for the tests. Following the explanation of the tests, the results will be tabulated and briefly discussed in sub-section 4.3.

4.2 Prototype Tests

This section discusses the different types of changes that will be made to the PDFs. These changes are what the prototype will attempt to detect. Sub-sections 4.2.1 - 4.2.5 explain the different changes, which are changes to the text, images, and metadata.

4.2.1 PDF Document Without Hashes

The simplest test that will be conducted on the PDF documents is using a PDF document that has not been protected by the prototype, i.e. it does not have any hashes stored.

4.2.2 Altering the Text

The most common form of change that will be made to a PDF document will be altering the text. We test the addition of text on a singular page, followed by testing the addition on multiple pages. Once we have tested the addition of text, we test the text alteration on single and multiple pages. The third type of test for altering text detection is removing text from a single page and multiple pages. Finally, to detect tampering with text, we will combine text addition, alteration, and removal in a combined test file.

4.2.3 Altering Images

The following significant change that we will assess the ability of the prototype to detect is alterations to an image. We test if the prototype can detect adding an image or multiple images on different pages. Following this, we assess the prototype's ability to detect alterations to an image on a page and, finally, its ability to detect the removal of images from a page or multiple pages.

4.2.4 Altering the Metadata

After we have tested the ability of the prototype to detect alterations to images, we assess its ability to detect changes to the metadata of the PDF document. We add elements, then we remove elements, and finally, alter metadata elements for the PDF document. The previously mentioned test cases are applied to singular and multiple elements of the PDF document.

4.2.5 Combination of All Alterations

The final test combines all of the types of alterations discussed in subsections 4.2.2 to 4.2.5.

4.3 Final Results

Tables 1 - 5 below tabulate the results of all the tests mentioned in subsection 4.2. All PDF documents used in the test cases described below were first protected by the prototype, then the alterations were made. The only exception to this protection is the first test, which explicitly tests for the absence of the hashes. The PDF documents were created using Microsoft Word and Adobe Acrobat. The PDF names with a * at the beginning denote that the PDF was created with Microsoft Word; those without were created with Adobe Acrobat. All changes to the text, images and metadata within a PDF were done using Adobe Acrobat. The file page objects will change depending on the changes made to the text and images within a PDF document.

Table 1: Experimentation Results: Not Protected

Test Name	PDF Name	Test Description
No Hashes	NoHash_hash.pdf	This test assesses the ability of the prototype to detect the absence of hashes in a PDF.
Result	<pre>Assessing: ./Test_PDFs/NoHash.pdf There was an error assessing the PDF: No hash values found in PDF Process Completed</pre>	

Table 1 shows that the prototype can detect when a PDF does not have the hashes stored within the file. It also indicates that it cannot assess such a file because no hashes are stored.

Table 2: Experimentation Results: Text Alteration

Test Name	PDF Name	Test Description
Single addition	* TextSA_hash.pdf	This test sees the addition of a singular line of text on the 2nd page of the PDF document.
Result	<pre>Assessing: ./Test_PDFs/TextSA_hash.pdf Hashes are not equal, alterations detected: Root Hashes are not equal, root object has been altered Info Hashes are not equal, metadata has changed Object Hashes are not equal for page: 2 Root Hashes are not equal for page: 2 Changes detected in the 0 th 256 bytes of the content stream Changes detected in the 1 th 256 bytes of the content stream</pre>	
Multiple addition	* TextMA_hash.pdf	This test sees the addition of multiple singular lines of text on the 2nd and 3rd page of the PDF document.
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Test Name	PDF Name	Test Description
Result		Assessing: ./Test_PDFs/TextMA_hash.pdf Hashes are not equal, alterations detected: Root Hashes are not equal, root object has been altered Info Hashes are not equal, metadata has changed Object Hashes are not equal for page: 2 Root Hashes are not equal for page: 2 Changes detected in the 0 th 256 bytes of the content stream Changes detected in the 1 th 256 bytes of the content stream Object Hashes are not equal for page: 3 Root Hashes are not equal for page: 3 Changes detected in the 0 th 256 bytes of the content stream Changes detected in the 1 th 256 bytes of the content stream
Single Update	* TextSU_hash.pdf	This test sees the update of a singular line of text on the 1st page of the PDF document.
Result		Assessing: ./Test_PDFs/TextSU_hash.pdf Hashes are not equal, alterations detected: Root Hashes are not equal, root object has been altered Info Hashes are not equal, metadata has changed Object Hashes are not equal for page: 1 Root Hashes are not equal for page: 1 Changes detected in the 0 th 256 bytes of the content stream Changes detected in the 1 th 256 bytes of the content stream Changes detected in the 2 th 256 bytes of the content stream Changes detected in the 3 th 256 bytes of the content stream
Multiple Update	* TextMU_hash.pdf	This test sees the update of multiple singular lines of text on the 1st and 2nd page of the PDF document.
Result		Assessing: ./Test_PDFs/TextMU_hash.pdf Hashes are not equal, alterations detected: Root Hashes are not equal, root object has been altered Info Hashes are not equal, metadata has changed Object Hashes are not equal for page: 1 Root Hashes are not equal for page: 1 Changes detected in the 0 th 256 bytes of the content stream Changes detected in the 1 th 256 bytes of the content stream Changes detected in the 2 th 256 bytes of the content stream Object Hashes are not equal for page: 2 Root Hashes are not equal for page: 2 Changes detected in the 0 th 256 bytes of the content stream Changes detected in the 1 th 256 bytes of the content stream Changes detected in the 2 th 256 bytes of the content stream
Single Deletion	* TextSD_hash.pdf	This test sees the deletion of a singular line of text on the 3rd page of the PDF document.
Result		Assessing: ./Test_PDFs/TextSD_hash.pdf Hashes are not equal, alterations detected: Root Hashes are not equal, root object has been altered Info Hashes are not equal, metadata has changed Object Hashes are not equal for page: 3 Root Hashes are not equal for page: 3 Changes detected in the 0 th 256 bytes of the content stream Changes detected in the 1 th 256 bytes of the content stream
Multiple Deletion	* TextMD_hash.pdf	This test sees the deletion of multiple singular lines of text on the 2nd and 3rd page of the PDF document.
Result		Assessing: ./Test_PDFs/TextMD_hash.pdf Hashes are not equal, alterations detected: Root Hashes are not equal, root object has been altered Info Hashes are not equal, metadata has changed Object Hashes are not equal for page: 2 Root Hashes are not equal for page: 2 Changes detected in the 0 th 256 bytes of the content stream Changes detected in the 1 th 256 bytes of the content stream Changes detected in the 2 th 256 bytes of the content stream Object Hashes are not equal for page: 3 Root Hashes are not equal for page: 3 Changes detected in the 0 th 256 bytes of the content stream Changes detected in the 1 th 256 bytes of the content stream
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Test Name	PDF Name	Test Description
Combination	* TextC_hash.pdf	This test sees a combination of addition, deletion, and updates throughout the PDF document.
Result		<pre> Assessing: ./Test_PDFs/TextC_hash.pdf Hashes are not equal, alterations detected: Root Hashes are not equal, root object has been altered Info Hashes are not equal, metadata has changed Object Hashes are not equal for page: 1 Root Hashes are not equal for page: 1 Changes detected in the 0 th 256 bytes of the content stream Changes detected in the 1 th 256 bytes of the content stream Changes detected in the 2 th 256 bytes of the content stream Changes detected in the 3 th 256 bytes of the content stream Object Hashes are not equal for page: 2 Root Hashes are not equal for page: 2 Changes detected in the 0 th 256 bytes of the content stream Changes detected in the 1 th 256 bytes of the content stream Changes detected in the 2 th 256 bytes of the content stream Object Hashes are not equal for page: 3 Root Hashes are not equal for page: 3 Changes detected in the 0 th 256 bytes of the content stream Changes detected in the 1 th 256 bytes of the content stream </pre>

Table 2 shows that the prototype can detect when the text in a PDF has been altered. Be this an addition, update, deletion, or a combination of all of the above. The messages displayed indicate which page or pages the text has been altered on and which parts of the content stream have been altered.

Table 3: Experimentation Results: Image Alteration

Test Name	PDF Name	Test Description
Single addition	ImageSA_hash.pdf	This test sees the addition of a singular image on the 2nd page of the PDF document.
Result		<pre> Assessing: ./Test_PDFs/ImageSA_hash.pdf Hashes are not equal, alterations detected: Root Hashes are not equal, root object has been altered Object Hashes are not equal for page: 2 Root Hashes are not equal for page: 2 Changes detected in the 0 th 256 bytes of the content stream </pre>
Multiple addition	ImageMA_hash.pdf	This test sees the addition of multiple images on the 2nd and 3rd page of the PDF document.
Result		<pre> Assessing: ./Test_PDFs/ImageMA_hash.pdf Hashes are not equal, alterations detected: Root Hashes are not equal, root object has been altered Object Hashes are not equal for page: 2 Root Hashes are not equal for page: 2 Changes detected in the 0 th 256 bytes of the content stream Object Hashes are not equal for page: 3 Root Hashes are not equal for page: 3 Changes detected in the 0 th 256 bytes of the content stream </pre>
Single Update	ImageSU_hash.pdf	This test sees the change of a singular image on the 1st page of the PDF document.
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Test Name	PDF Name	Test Description
Result		Assessing: ./Test_PDFs/ImageSU_hash.pdf Hashes are not equal, alterations detected: Root Hashes are not equal, root object has been altered Object Hashes are not equal for page: 1 Root Hashes are not equal for page: 1 Changes detected in the 0 th 256 bytes of the content stream
Multiple Update	ImageMU_hash.pdf	This test sees the change of multiple images on the 1st and 3rd page of the PDF document.
Result		Assessing: ./Test_PDFs/ImageMU_hash.pdf Hashes are not equal, alterations detected: Root Hashes are not equal, root object has been altered Info Hashes are not equal, metadata has changed Object Hashes are not equal for page: 1 Root Hashes are not equal for page: 1 Changes detected in the 0 th 256 bytes of the content stream Changes detected in the 1 th 256 bytes of the content stream Changes detected in the 2 th 256 bytes of the content stream Changes detected in the 3 th 256 bytes of the content stream Object Hashes are not equal for page: 3 Root Hashes are not equal for page: 3 Changes detected in the 0 th 256 bytes of the content stream Changes detected in the 1 th 256 bytes of the content stream
Single Deletion	ImageSD_hash.pdf	This test sees the deletion of a singular image on the 3rd page of the PDF document.
Result		Assessing: ./Test_PDFs/ImageSD_hash.pdf Hashes are not equal, alterations detected: Root Hashes are not equal, root object has been altered Object Hashes are not equal for page: 3 Root Hashes are not equal for page: 3 Changes detected in the 0 th 256 bytes of the content stream
Multiple Deletion	ImageMD_hash.pdf	This test sees the deletion of multiple images on the 2nd and 3rd page of the PDF document.
Result		Assessing: ./Test_PDFs/ImageMD_hash.pdf Hashes are not equal, alterations detected: Root Hashes are not equal, root object has been altered Info Hashes are not equal, metadata has changed Object Hashes are not equal for page: 1 Root Hashes are not equal for page: 1 Changes detected in the 0 th 256 bytes of the content stream Changes detected in the 1 th 256 bytes of the content stream Changes detected in the 2 th 256 bytes of the content stream Object Hashes are not equal for page: 2 Root Hashes are not equal for page: 2 Changes detected in the 0 th 256 bytes of the content stream Changes detected in the 1 th 256 bytes of the content stream Object Hashes are not equal for page: 3 Root Hashes are not equal for page: 3 Changes detected in the 0 th 256 bytes of the content stream Changes detected in the 1 th 256 bytes of the content stream
Combination	ImageC_hash.pdf	This test sees a combination of addition, deletion, and updates throughout the PDF document.
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Test Name	PDF Name	Test Description
Result		<pre> Assessing: ./Test_PDFs/ImageC_hash.pdf Hashes are not equal, alterations detected: Root Hashes are not equal, root object has been altered Info Hashes are not equal, metadata has changed Object Hashes are not equal for page: 1 Root Hashes are not equal for page: 1 Changes detected in the 0 th 256 bytes of the content stream Changes detected in the 1 th 256 bytes of the content stream Changes detected in the 2 th 256 bytes of the content stream Object Hashes are not equal for page: 2 Root Hashes are not equal for page: 2 Changes detected in the 0 th 256 bytes of the content stream Changes detected in the 1 th 256 bytes of the content stream Object Hashes are not equal for page: 3 Root Hashes are not equal for page: 3 Changes detected in the 0 th 256 bytes of the content stream Changes detected in the 1 th 256 bytes of the content stream </pre>

Table 3 successfully detects when an image or images in a PDF have been altered, whether it is an addition, update, deletion, or a combination of all of the above. The messages displayed indicate which page or pages the text has been altered on and which parts of the content stream have been altered.

Table 4: Experimentation Results: Metadata Alteration

Test Name	PDF Name	Test Description
Single Update	* MetaSU_hash.pdf	This test sees the addition of a singular meta-data element in a PDF document.
Result		<pre> Assessing: ./Test_PDFs/MetaSU_hash.pdf Hashes are not equal, alterations detected: Info Hashes are not equal, metadata has changed </pre>
Multiple Update	* MetaMU_hash.pdf	This test sees the update of multiple meta-data elements in the PDF document.
Result		<pre> Assessing: ./Test_PDFs/MetaMU_hash.pdf Hashes are not equal, alterations detected: Info Hashes are not equal, metadata has changed </pre>
Single Deletion	* MetaSD_hash.pdf	This test sees the addition of a singular meta-data element in a PDF document.
Result		<pre> Assessing: ./Test_PDFs/MetaSD_hash.pdf Hashes are not equal, alterations detected: Info Hashes are not equal, metadata has changed </pre>
Multiple Deletion	* MetaMD_hash.pdf	This test sees the deletion of multiple meta-data elements in the PDF document.
Result		<pre> Assessing: ./Test_PDFs/MetaMD_hash.pdf Hashes are not equal, alterations detected: Info Hashes are not equal, metadata has changed </pre>
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Test Name	PDF Name	Test Description
Combination	* MetaC_hash.pdf	This test sees a combination of addition, deletion, and updates of the metadata elements in the PDF document.
Result		Assessing: ./Test_PDFs/MetaC_hash.pdf Hashes are not equal, alterations detected: Info Hashes are not equal, metadata has changed

Table 4 shows that the prototype can detect tampering with the metadata of a PDF document. The message displayed indicates that the metadata has been tampered with.

Table 5: Experimentation Results: Combined Test

Test Name	PDF Name	Test Description
Combination Of All alterations		
Combination All	* CAll_hash.pdf	This test sees a combination of the above alterations to a PDF document.
Result		Assessing: ./Test_PDFs/CAll_hash.pdf Hashes are not equal, alterations detected: Root Hashes are not equal, root object has been altered Info Hashes are not equal, metadata has changed Changes detected in the 1 th 256 bytes of the content stream Changes detected in the 2 th 256 bytes of the content stream Changes detected in the 3 th 256 bytes of the content stream Object Hashes are not equal for page: 2 Root Hashes are not equal for page: 2 Changes detected in the 0 th 256 bytes of the content stream Changes detected in the 1 th 256 bytes of the content stream Changes detected in the 2 th 256 bytes of the content stream Changes detected in the 3 th 256 bytes of the content stream Object Hashes are not equal for page: 3 Root Hashes are not equal for page: 3 Changes detected in the 0 th 256 bytes of the content stream

Table 5 shows that the prototype can detect all the tampering tested for in the previous experiments when combined into a single file tampering. The messages displayed indicate that the metadata and the content of the pages have been tampered with.

The following section concludes the chapter.

4.4 Conclusion

This chapter discussed the tests that the prototype was used in. It elaborated on the types of changes that would be made, such as text, image and metadata. The different tests and their results are displayed in tables 1 - 5. The following chapter will critically analyse the prototype and its ability to detect tampering or forgery. Some consideration will be given to the shortcomings of the prototype.